

LINUX LAB-1

Que-1

Que 1: Create a LabFiles directory and perform create/copy/move/rename/delete operations using touch, cp, mv, rm, unlink, ls, stat.

Ans 1:

Theory

Linux provides several commands for managing files and directories. mkdir creates a directory, while touch creates empty files. ls displays directory contents. cp copies files, whereas mv is used to move or rename files. stat displays detailed file metadata such as size, permissions and timestamps. rm deletes files, while unlink removes a single file or hard link. These commands are used to perform basic file management operations in Linux.

Result

The LabFiles directory was created successfully, and file creation, copying, renaming, metadata checking, deletion and hard link removal operations were performed successfully using Linux commands.

```

y:/home# mkdir LabFiles && cd LabFiles
y:/home/LabFiles# touch file1.txt file2.txt
y:/home/LabFiles# ls -l
- 1 root root 0 Sep 19 16:57 file1.txt
- 1 root root 0 Sep 19 16:57 file2.txt
y:/home/LabFiles# cp file1.txt file1_copy.txt
y:/home/LabFiles# ls -l
- 1 root root 0 Sep 19 16:57 file1.txt
- 1 root root 0 Sep 19 16:57 file1_copy.txt
- 1 root root 0 Sep 19 16:57 file2.txt
y:/home/LabFiles# mv file2.txt file2_renamed.txt
y:/home/LabFiles# ls -l
- 1 root root 0 Sep 19 16:57 file1.txt
- 1 root root 0 Sep 19 16:57 file1_copy.txt
- 1 root root 0 Sep 19 16:57 file2_renamed.txt
y:/home/LabFiles# stat file1_copy.txt
file1_copy.txt
      Blocks: 0          IO Block: 4096   regular empty file
,48   Inode: 41904       Links: 1
0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
026-09-19 16:57:57.363262268 +0000
026-09-19 16:57:57.363262268 +0000
026-09-19 16:57:57.363262268 +0000
026-09-19 16:57:57.363262268 +0000
y:/home/LabFiles# rm file1_copy.txt
y:/home/LabFiles# ls -l
- 1 root root 0 Sep 19 16:57 file1.txt
- 1 root root 0 Sep 19 16:57 file2_renamed.txt
y:/home/LabFiles# ln file2_renamed.txt file2_hardlink.txt
y:/home/LabFiles# unlink file2_hardlink.txt
y:/home/LabFiles# ls -l
- 1 root root 0 Sep 19 16:57 file1.txt
- 1 root root 0 Sep 19 16:57 file2_renamed.txt
y:/home/LabFiles# ls -l

```